

What is a 'robot' and what makes it autonomous?

IEEE 1872:

Robot - An **agentive device** in a broad sense, purposed to **act in the physical world in order to accomplish one or more tasks**. In some cases, the actions of a robot might be subordinated to actions of other agents, such as software agents (bots) or humans. A robot is composed of suitable **mechanical and electronic parts**. Robots might form social groups, where they interact to achieve a common goal. A robot (or a group of robots) can form robotic systems together with special environments geared to facilitate their work

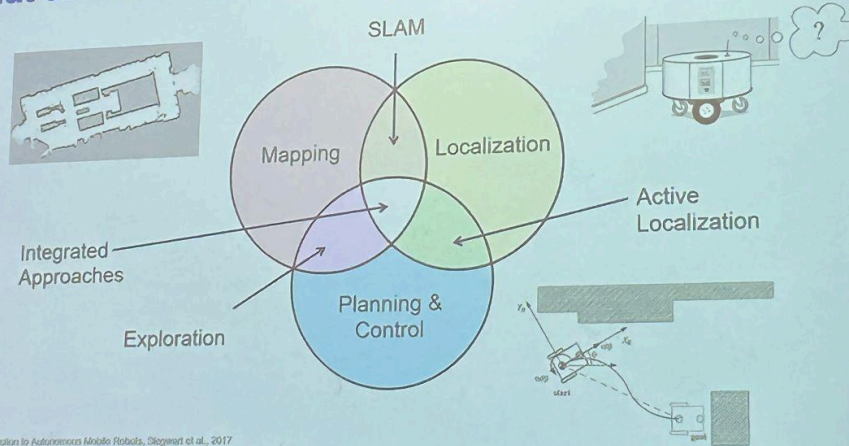
ISO 8373:2021:

Robot - programmed **actuated mechanism** with a **degree of autonomy** to perform locomotion, manipulation or positioning; A robot includes the **control system**; Examples of mechanical structure of robots are **manipulator**, **mobile platform** and **wearable robot**.

Beer et al. (Journal of Human-Robot Interaction):

Autonomy - The extent to which a robot can **sense the environment**, **plan based on that environment**, and **act upon that environment**, with the intent of reaching some goal (either given to or created by the robot) without external control.

What are common concepts?



Integrates from Introduction to Autonomous Mobile Robots, Skogman et al., 2017

Assessments

Quizzes (20%) ?

10 quizzes in total (2% each)
Short questions on lecture content
Posted after lecture, due before next
Designed to take ~ 30 minutes
Completed **individually**
Access via D2L Brightspace

Assignments (50%)

4 assignments in total (2x10%, 2x15%)
Implement core concepts on real robots
~2-3 weeks per assignment
Lab time dedicated to working on them
Completed in **groups** of 3-4
Submission via D2L Brightspace & demo showcase during labs

Final Exam (30%)

Written exam at end of term
Tests fundamentals and concepts
Questions similar to quizzes
2-3 hours in length
Completed **individually**
Allowed: one double-sided handwritten notes page

Assignments

Assignment 1: Mobile Robot Motion

Task Description:
Robot to drive along predefined paths in open-loop.

Assignment 2: Obstacle Avoidance

Task Description:
Robot to drive continuously while avoiding obstacles.

Assignment 3: Closed-Loop Control

Task Description:
Robot to follow a wall at a given distance and detects corners to turn accordingly.

Assignment 4: Pick-Up and Delivery

Task Description:
Robot to follow a wall while tracking its covered distance. At certain distances robot will seek out and follow lines orthogonal to the wall to pick-up and deliver objects between locations.

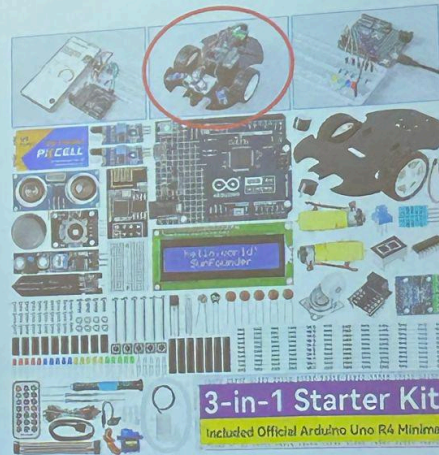
Robot Hardware Component

Robot Kit

- Strictly required for labs and assignments
- Kit is shared between your group for assignments of 3-4 students
- Free to design your robot however you want to fulfill the tasks at hand!

Recommended Option

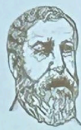
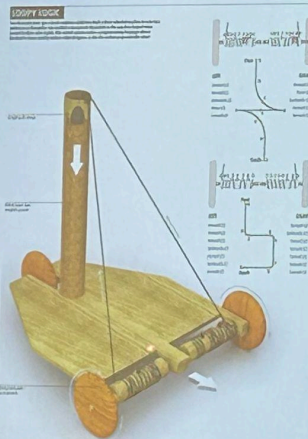
- SunFounder 3 in 1 Ultimate Starter Kit (~100\$ + tax):
<https://www.sunfounder.com/products/sunfounder-3-in-1-ultimate-starter-kit-with-original-arduino-uno-r4-minima?pos=4&sid=0d97d64bf&ss=r>
- Additional wires, pegboards, and sensors might be required for later assignments
- Easy to assemble, online guides for setup, includes enough to get started (and some extra parts we won't use)
- Cost saving option: Buy parts individually (but might require more work)
- List of recommended components on Brightspace
- Note: Please purchase the hardware early to account for shipping delays!



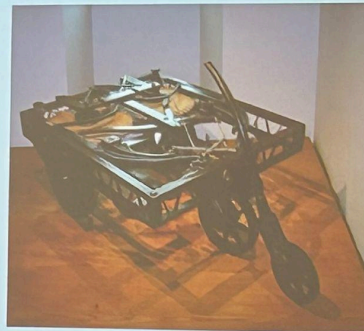
History of Mobile Robotics

Some of the Earliest Mobile Robots

Hero of Alexandria - 70 AD

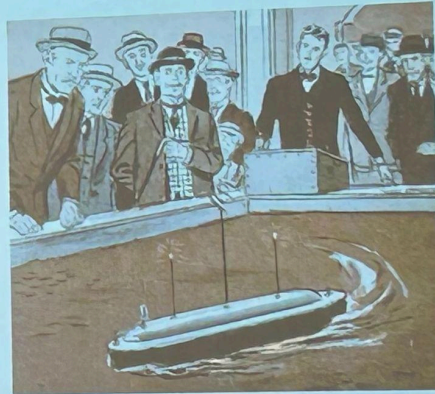
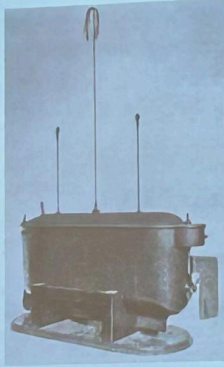


Leonardo Da Vinci - 1478



First programmable mobile robots

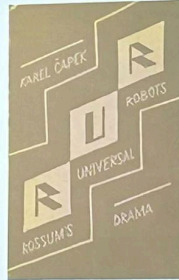
1898: Nikola Tesla's Radio-Controlled Boat



U.S. patent 613,809
*Method of an Apparatus for Controlling
Mechanism of Moving Vehicle or Vehicles*

First teleoperation
of a mobile robot

1920: Introduction of the Word 'Robot'

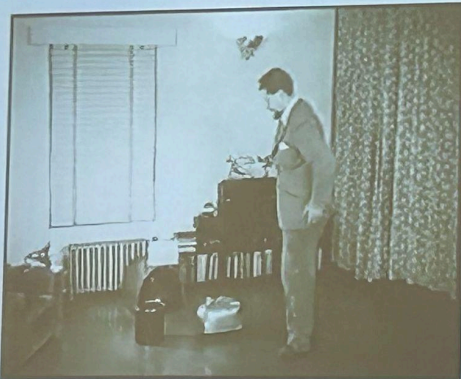


Rossum's Universal Robots (Rossumovi Univerzální Roboti)

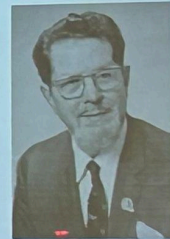
- Czech play by Karel Čapek
- Introduced the word 'robot' to the English language and to science-fiction
- In Czech, 'robota' means forced labour and is derived from 'rab', meaning 'slave'.

First use of the
word 'robot'

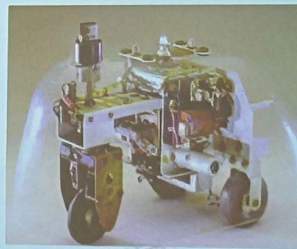
1948/49: Machina Speculatrix



- William Grey Walter built two *tortoise* robots
- Their electronics were fully analogous
- No fixed behaviour, but based on reflexes
- Could avoid obstacles based on touch sensors
- Used the principle of phototaxis to seek light sources, allowing them to return to a charging station

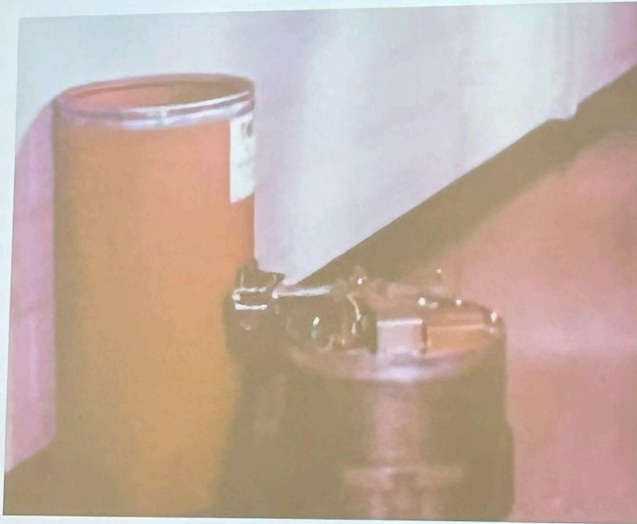


William Grey
Walter



First mobile robot with
autonomous behaviour

1960s: John Hopkins Beast



- Built at the Johns Hopkins University Applied Physics Laboratory
- Consisted of several transistors controlling analog voltages
- Initially relied only on touch sensors
- Later version included sonar sensors as well as photocell optics to navigate white halls and detect black outlets

First robot with deliberate and coordinated actions

1969: Mowbot



We sell leisure...we just call it a lawnmower



- Autonomy: 3 hours
- Coverage: 280 m²
- Cost: \$800 USD (≈ \$7,000 USD in 2025)

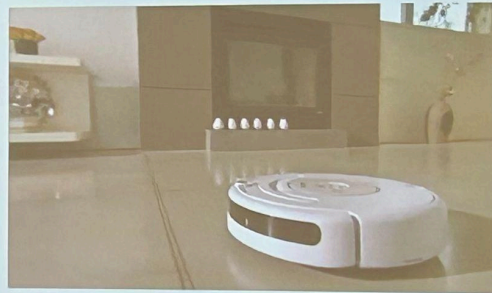
First outdoor mobile robot and first field robotics application

1990 - present: iRobot

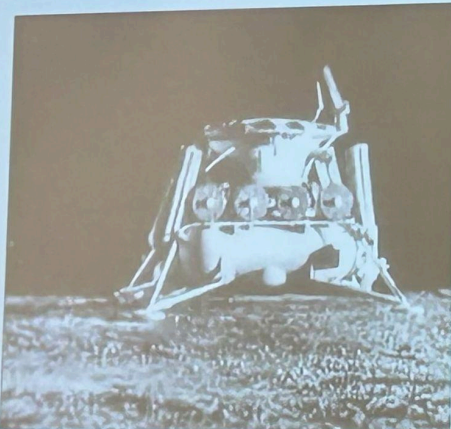


First mobile robot with commercial success

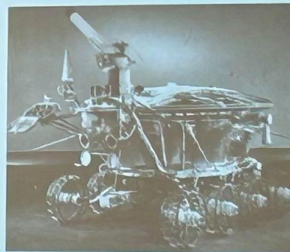
- 1990: Rodney Brooks founded iRobot
- 1999: First commercial success with PackBot
- Mainly used for bomb disposal & first to enter the damaged Fukushima nuclear reactor in 2011
- 2002: Started selling Roomba vacuum-cleaner robot
- To date, more than 40 million home robots sold worldwide



1970-73: Lunokhod



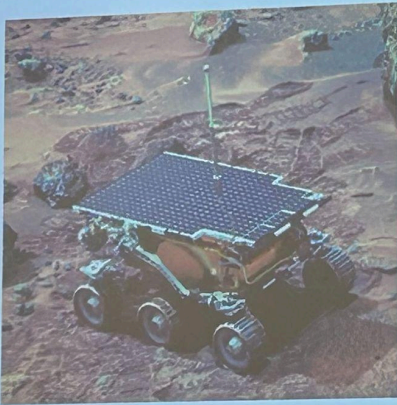
- USSR landed two teleoperated rovers on the Moon
- Lunokhod 1 (1971) travelled 11 km
- Lunokhod 2 (1973) travelled 37 km



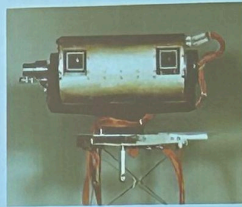
First mobile robot on another planetary surface



1997: Sojourner

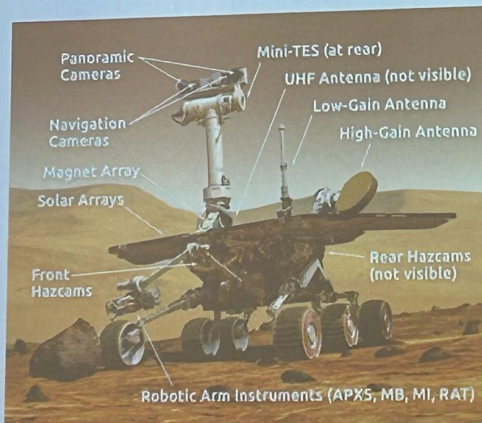


- NASA landed a small rover on Mars in 1997
- Sojourner had simple sensing, but mostly relied on script commands based on images taken from the lander
- Traveled ~100 m, never more than 12 m away from lander
- Survived 83 Martian sols, sent back 550 images, and visited 16 sites of interest



First mobile robot on Mars

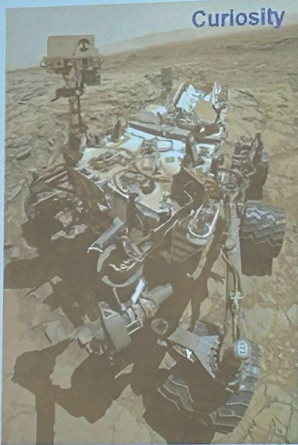
2004 - 2018: Spirit & Opportunity



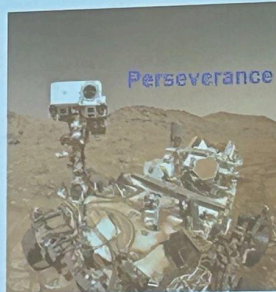
- **2004:** NASA lands two rovers Spirit and Opportunity on Mars, supposed to last 90 days
- **2011:** Last transmission to Spirit, stuck in sand trap
- **2014:** Opportunity had driven the accumulated distance of 40 km, further than any other vehicle on a world other than Earth
- **2018:** Opportunity caught in dust storm, lost contact forever

First mobile robot with on-board autonomy on another planetary surface

2012 – present: Curiosity & Perseverance



Curiosity



Perseverance

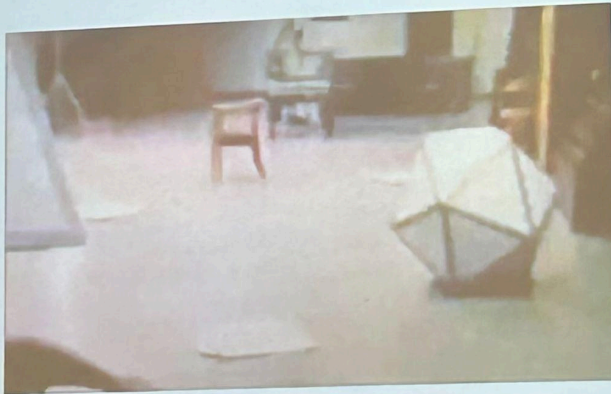


Ingenuity

- **2012:** NASA lands rover Curiosity near Gale Crater on Mars to search for past life
- **2013:** Confirms Mars once had conditions suitable for life
- **2021:** NASA lands rover Perseverance in Jezero Crater on Mars. It collects rock samples for return to Earth and employs helicopter Ingenuity.
- **2022:** Findings indicate past water activity on Mars.
- **2024:** Ingenuity's rotor blades break during the landing of the 72nd flight. Huge success, since it was intended to only make five flights.

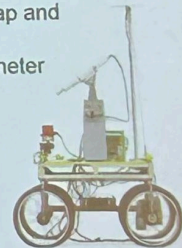
First helicopter to fly on another planet

1960/70s: The Stanford Cart

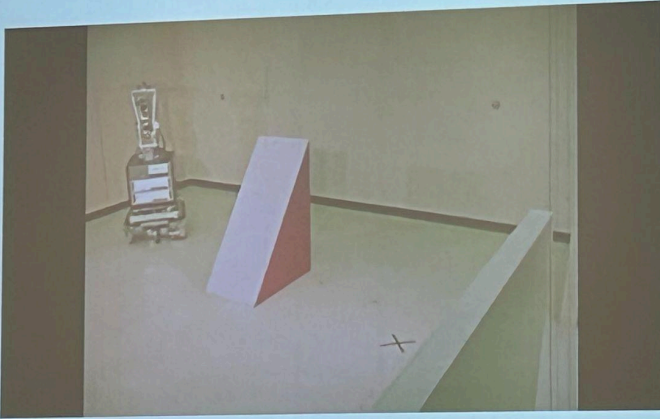


First mobile robot to feature 3D vision and autonomous mapping/planning

- **1960/61:** Built by James Adams to study whether a rover could be teleoperated on the moon
- **1962/63:** Paul Braisted added a predictive display, increased the speed at which the vehicle could be teleoperated
- **1966/71:** Rodney Schmidt added autonomous line following under controlled lighting
- **1971-1979:** Hans Moravec and Victor Scheiman turned the cart's camera into a slider, making it the first robot with 3D vision. Cart had the ability to build a map and autonomously plan routes. Its workflow involved driving 1 meter at a time, then spending about 15 minutes computing the next motion step.



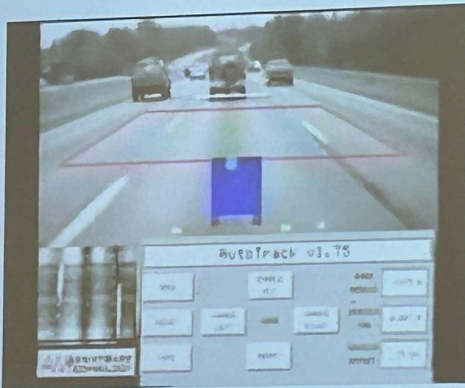
1966-1972: Shakey



- Built at the Stanford Research Institute under Charles Rosen
- Featured TV camera, triangulating range sensor, bump sensors, computers, radio communications
- Algorithms for perception, world-modelling, and action. Capable of breaking down commands into low-level actions.

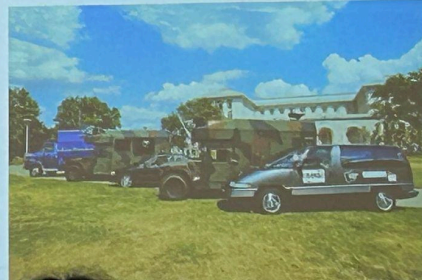
First mobile robot to reason about its actions

1986-1997: Navlab



First robot to drive across the US

- Developed by CMU starting in 1986
- Navlab 1 drove 32 km/h on the road
- Navlab 2 also drove off-road at 9.7 km/h
- Navlab 1 & 2 were large vehicles due to massive computing hardware onboard
- Navlab 5 was much simpler, featuring a forward-looking camera and using a neural network (ALVINN). Made the first drive across the US (Pittsburgh to LA) with 98.2% autonomy rate and up to 112 km without human input.



- NOTE: Self-driving

2004 - present: Self-Driving Cars

DARPA Challenges



CMU Sandstorm (2004)



CMU Boss (2007)

Google Self-Driving (2009)



Current Landscape



Waymo (since 2016)



Tesla (since 2016)

2003: Kiva Robotics



- Kiva Systems started a warehouse picking system in 2003 and by 2009 was the fastest-growing company in the US.
- Uses hundreds of robots to bring shelves of items to a person in a fixed workstation who then picks the items and puts them in a box.
- Bought by Amazon in 2012 for \$775M cash

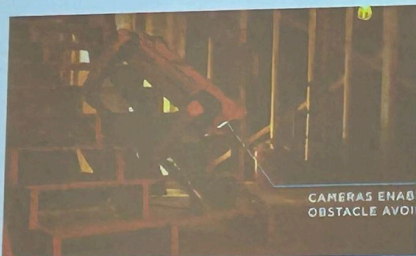


2005: BigDog

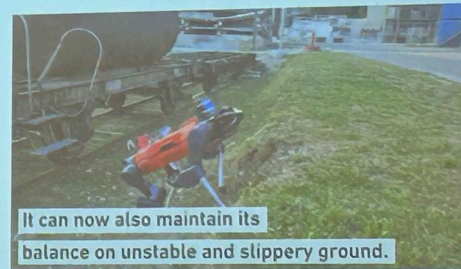


- BigDog is a quadruped robot built by Boston Dynamics to serve as a soldier's pack mule.
- It can travel 6.4 km/h carrying 150 kg, and walk on very rough terrain including 35 degree slopes.
- Balance control and navigation is autonomous using a suite of 50 different sensors.

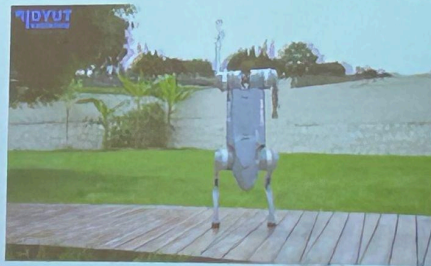
Present: Modern Quadrupeds



Spot (Boston Dynamics)



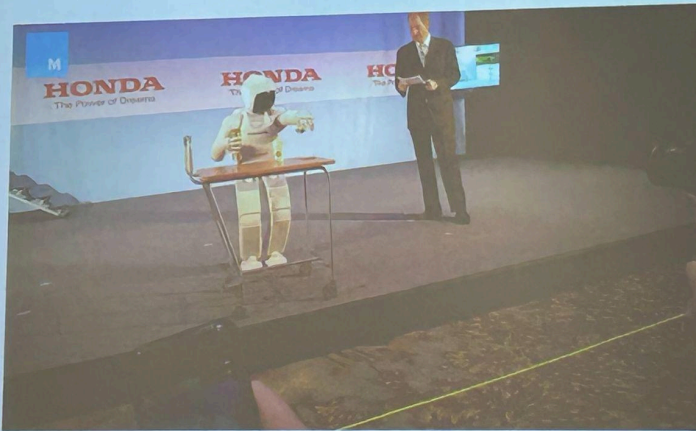
ANYmal (ANYbotics)



(Unitree)

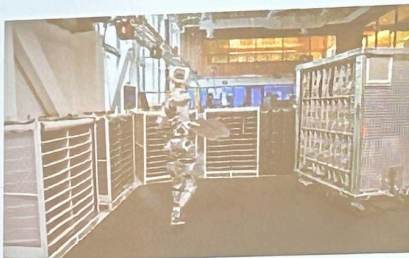
* Go2 (Unitree)

2000: ASIMO

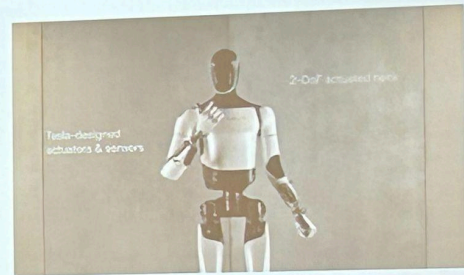


- HONDA developed the now-famous Asimo humanoid robot platform in secret, spending over \$100 million on the prototype revealed in 2000.
- Able to run, interact with people, perform tasks, serving as a research platform for HONDA.
- Development ended in 2022 to focus on more practical applications.

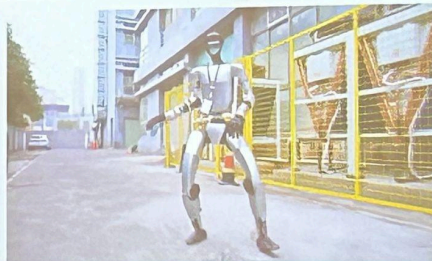
Present: Modern Humanoids



Atlas (Boston Dynamics)



Optimus (Tesla)



G1 (Unitree)